ASSIGNMENT 1 – CO1 | TASK 6

Control Structures and Completed Fare Code

Scenario: Namma Metro Smart-Card Fare System

Objective / explanation

A switch selects the base fare. Integer arithmetic adds a 20% surcharge during peak hours. The journey is allowed only when at least Rs 50 remains.

C++ source code

```
#include &lt;iomanip&gt;
#include &lt;iostream&gt;
using namespace std;
enum class Zone { Inner, Middle, Outer };
constexpr int BASE[] = {1000, 2000, 3000};
constexpr int PEAK_PERCENT = 20;
constexpr long long MIN_BALANCE = 5000;

long long fare(Zone z, bool peak) {
    long long base = 0;
    switch (z) {
        case Zone::Inner: base = BASE[0]; break;
        case Zone::Middle: base = BASE[1]; break;
        case Zone::Outer: base = BASE[2]; break;
        default: return -1;
    }
    if (peak) base += base * PEAK_PERCENT / 100;
    return base;
}
bool canTravel(long long balance, long long f) {
    return f &gt;= 0 &amp;&amp; balance &gt;= f &amp;&amp; balance - f &gt;= MIN_BALANCE;
}
int main() {
    long long balance = 10000;
    struct Trip { Zone zone; bool peak; };
    Trip trips[] = {{Zone::Inner,false},{Zone::Outer,true},{Zone::Middle,false}};
    for (const auto&amp; t : trips) {
        long long f = fare(t.zone, t.peak);
        if (canTravel(balance, f)) {
            balance -= f;
            cout &lt;&lt; "Fare Rs " &lt;&lt; f/100 &lt;&lt; '.' &lt;&lt; setw(2)
                 &lt;&lt; setfill('0') &lt;&lt; f%100 &lt;&lt; setfill(' ');
        } else cout &lt;&lt; "Denied (low balance)";
        cout &lt;&lt; " balance Rs " &lt;&lt; balance/100 &lt;&lt; '\n';
    }
}
```

Compilation

```
g++ -std=c++17 -Wall -Wextra -pedantic main.cpp -o program
```

Actual captured program output

```
Fare Rs 10.00 balance Rs 90
Fare Rs 36.00 balance Rs 54
Denied (low balance) balance Rs 54
```

Output screenshot

Analysis / complexity

The range-based for loop processes each trip. switch handles the zone and default rejects an invalid value. Every fare and balance uses integer paise.

Evidence note: Output shown is captured by executing this source in the generation environment. Recompile and run on your own system, and use your own screenshots if your instructor requires evidence from your computer.